



Pharm*Achieve*

GOUT

Qualifying Exam Preparatory Course

COVERED IN THIS PRESENTATION...

**Providing Care:
Clinical Care**

**Providing Care:
Distribution**

**Knowledge
& Expertise**

**Communication
& Collaboration**

**Leadership &
Stewardship**

Professionalism

LEGEND

UA Uric Acid

PG Prostaglandin

BID Twice Daily

BP Blood Pressure

GI Gastrointestinal

CNS Central Nervous System

PUD Peptic Ulcer Disease

CHF Congestive Heart Failure

CAD Coronary Artery Disease

IBD Inflammatory Bowel Disease

HTN Hypertension

CI Contraindicated

NSAID Non-steroidal Anti-inflammatory Drugs

IV Intravenous

IM Intramuscular

PO Oral

N/V/D Nausea/Vomiting/Diarrhea

D/C Discontinue

WBC White Blood Cells

LFT Liver Function Tests

CV Cardiovascular

ADR Adverse Drug Reaction

ULT Urate Lowering Therapy

URTI Upper Respiratory Tract Infection

WHAT IS GOUT?

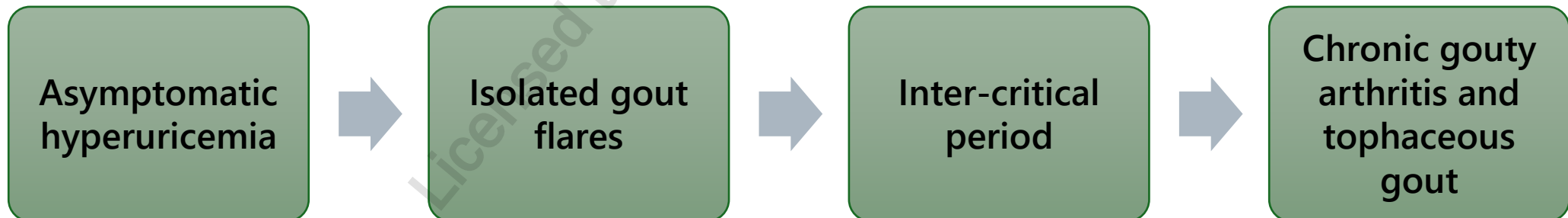
- Metabolic disease marked by monosodium urate monohydrate (MSU) crystal deposits in joints, soft tissues and renal tissue
- These deposits can cause gouty arthritis, tophi, nephropathy or uric acid nephrolithiasis
- Tophi/Tophaceous gout are nodules in soft tissue consisting of uric acid crystals. Strongest risk factor for tophi development is elevated serum urate level.

What is uric acid and what is its role in gout?

- Uric acid is the end-product of purine metabolism
- Increased serum uric acid levels lead to crystallization
- When these urate crystals deposit in joints, they cause inflammation and extreme pain
- Gout is a result of increased uric acid due to either **over-production** or **under-excretion of uric acid**

CLINICAL PRESENTATION

- Abrupt onset of excruciating pain and inflammation that tends to affect one joint initially, usually the large toe (podagra), causing the patient to limp while walking due to pain
 - Subsequent attacks may involve many joints (polyarticular), such as insteps, ankles, heels, knees, wrists, hands and elbows
- Night awakenings may occur due to increased pain at night when feet are elevated
- **Key symptoms:** pain, swelling, redness (dusky-red or cyanotic joint), warmth in the joint, stiffness in the joint
- **4 stages**



RISK FACTORS

- Males (>40 years of age)
- Increased alcohol intake (beer increases the risk 2-fold over liquor)
- Purine-rich diet from red meats, organ meats, sardines and shellfish
- Excessive consumption of asparagus, spinach, beans, peas, lentils, oatmeal, cauliflower and mushrooms
- Obesity (BMI >30 kg/m²)
- Renal failure
- Metabolic syndrome (cardiovascular disease, diabetes mellitus, obesity, hypertension, dyslipidemia)
- Lead intoxication
- Trauma
- Drug causes

DRUG-RELATED CAUSES

Memory Aid: C²ANT LEA³P

Cyclosporine: increases uric acid reabsorption in kidneys

Chemotherapy: kills cells, cell components (like purines) are released, which increases uric acid levels

Alcohol: contains yeast which has a high purine content, thus increasing uric acid levels

Niacin: Competes with uric acid for excretion into urine

Thiazide diuretics: Increase uric acid reabsorption in kidneys (dose dependent)

Loop diuretics: Increase uric acid reabsorption in kidneys (dose dependent)

Ethambutol decreases uric acid clearance

ASA (or salicylates): may worsen gout at low doses

- At low dose (< 2g/d): ASA increases serum uric acid levels by inhibiting excretion of uric acid in the urine.
 - However, with the use of very low ASA doses of 81 mg, benefits outweigh risks
- At higher dose (> 2g/d): ASA decreases uric acid levels by inhibiting both the excretion and reabsorption of uric acid within the kidneys. However, the inhibition of reabsorption of the uric acid predominates, therefore, the net effect is decreased serum levels of uric acid, with increased uric acid levels within the urine (uricosuria)

Allopurinol: can precipitate attack by mobilizing uric acid stores

ACEI: can reduce renal function thereby reducing clearance of uric acid

Pyrazinamide reduces uric acid clearance

GOALS OF THERAPY

- ✓ Eliminate signs and symptoms of gout
- ✓ Prevent future episodes
- ✓ Minimize adverse reactions due to therapy
- ✓ Use appropriate analgesic and anti-inflammatory agents at correct dose and frequency

NON-PHARMACOLOGICAL MEASURES

Memory Aid: WADI

W

Weight loss

A

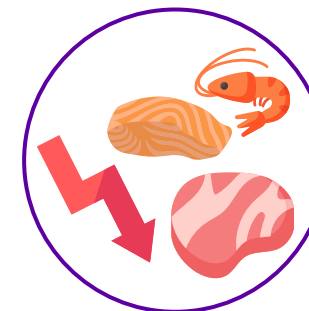
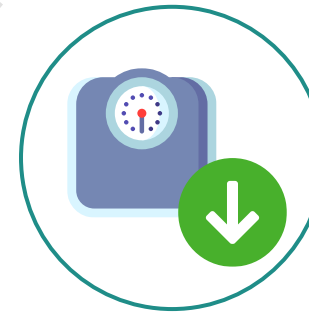
Limit beer & alcohol

D

Dietary changes to reduce fructose & purine consumption

I

Ice area during an acute attack



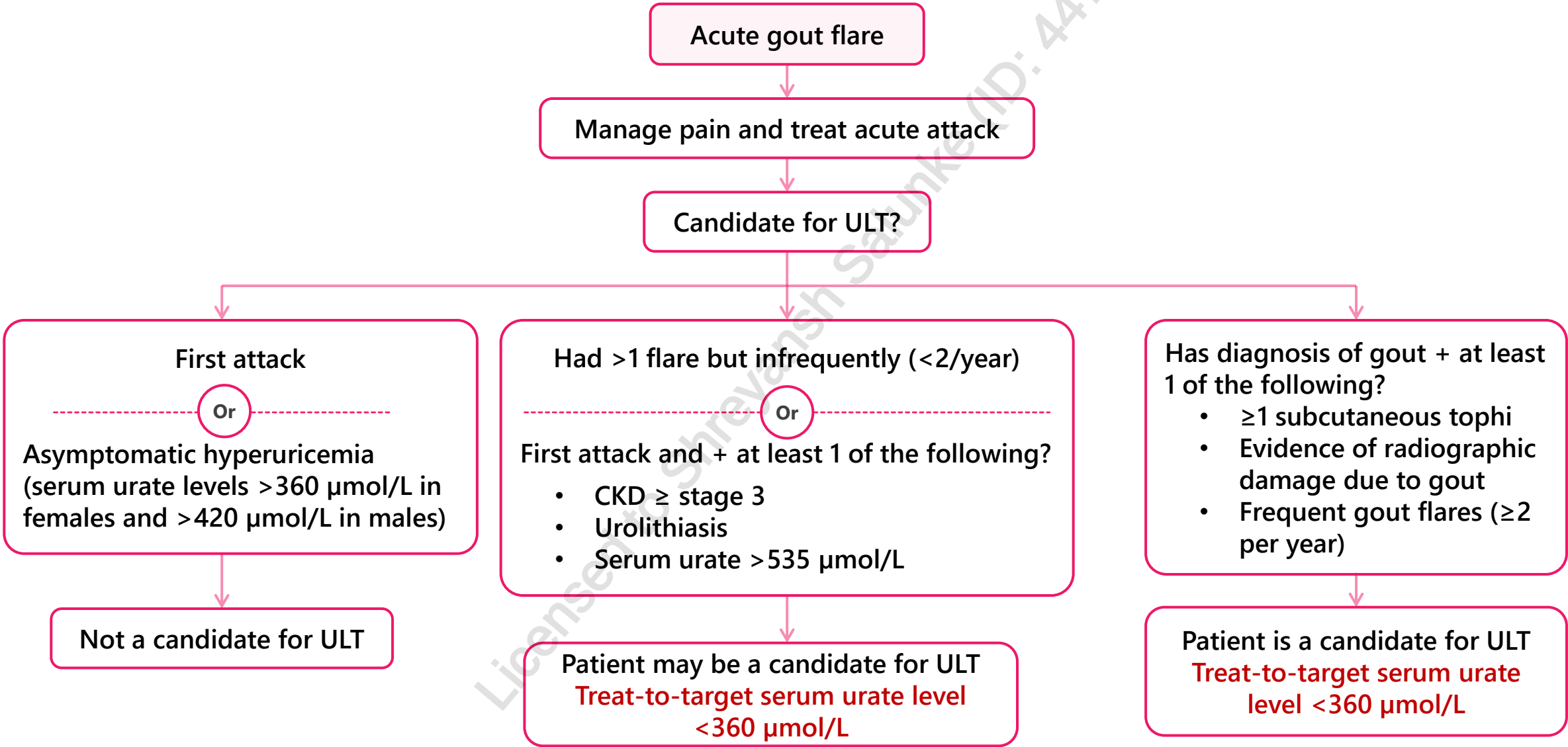
DIETARY MANAGEMENT

Foods and drinks to AVOID:

- High doses of niacin (seaweed, tuna, turkey, supplements)
- Organ meats high in purine, such as liver, kidney, sweetbreads
- Purine-rich seafood e.g. sardines, shellfish
- Foods and drinks high in fructose, corn syrup, sodium, table sugar (sucrose)
- Alcohol



CANDIDACY FOR ULT ALGORITHM



PHARMACOLOGICAL MEASURES - ACUTE CARE

PAIN AND INFLAMMATION

- NSAIDs, corticosteroids and colchicine are **appropriate first-line options**
- Consider patient-specific factors (concomitant medications, comorbidities, dosage form acceptability, etc.) when choosing an option
 - For example, patients with an increased risk of experiencing adverse effects with NSAIDs, colchicine and oral corticosteroids may benefit from injectable triamcinolone, especially if only one joint is affected
- Should be started immediately to reduce inflammation and alleviate symptoms
- Duration of acute therapy varies based on symptom resolution. Therapy may be tapered down and thus longer in patients who are on corticosteroids to prevent risk of rebound flares

PHARMACOLOGICAL MEASURES - ACUTE CARE

PAIN AND INFLAMMATION

NSAIDs (indomethacin, celecoxib, ibuprofen, naproxen, meloxicam)	• All non-salicylate NSAIDs are effective • Caution in patients with cardiovascular disease, renal disease, gastrointestinal disease (e.g., history of ulcers). May need addition of proton pump inhibitor (PPI) for gastric protection
Corticosteroids, systemic (prednisone) or intraarticular (triamcinolone acetonide, methylprednisolone acetate)	• Preferred in renal dysfunction • Monitor for steroid-related adverse effects like hypertension, hyperglycemia, etc. • IM/intra-articular formulations also available
Colchicine	• Metabolized by CYP3A4; avoid in severe renal impairment ($\text{CrCl} < 30 \text{ mL/min}$) • Monitor for diarrhea and blood dyscrasias • Dose: • Gout flare: 1.2 mg when flare begins, then 0.6 mg 1 hr later (max 1.8 mg over this 1 h period) • May repeat after 3 days • Wait 12 hours before restarting prophylactic dose of 0.6 mg once daily or BID

PHARMACOLOGICAL MEASURES - CHRONIC PROPHYLAXIS

URATE LOWERING THERAPY (ULT) OVERVIEW

- **Initiate** ULT if patient has a diagnosis of gout and any of the following:
 - ≥ 1 subcutaneous tophi
 - Evidence of radiographic damage due to gout
 - Frequent gout flares (≥ 2 /year)
- **May initiate**
 - ULT if patient has had >1 flare but infrequently (<2 /year)
 - Patients with first gout attack + any of: CKD $\geq$ stage 3, urolithiasis or serum urate >535 $\mu\text{mol/L}$
 - ULT during gout flare rather than after it has resolved, to capitalize on patient motivation to start therapy and reduce follow-up visits. Initiating ULT may worsen the flare due to uric acid mobilization. Need to consider patient-specific factors, the patient should be part of the decision
- **Do NOT initiate**
 - Asymptomatic hyperuricemia (serum urate levels >360 in females and >420 $\mu\text{mol/L}$ in males)
 - ULT in patients with first gout attack who do *not* fit in the above categories (i.e. ULT **may be initiated** for those who also have CKD $\geq$ stage 3, urolithiasis or serum urate >535 $\mu\text{mol/L}$)

PHARMACOLOGICAL MEASURES - CHRONIC PROPHYLAXIS

URATE LOWERING THERAPY (ULT) OVERVIEW

- If not already on anti-inflammatory therapy, **initiate** it as prophylaxis therapy when starting ULT. This is called the inter-critical period. **Use for 3-6 months**. May need to use for longer if patient continues experiencing flares.
- Use treat-to-target strategy: titrate dose of ULT to achieve **target serum urate <360 µmol/L**. Titration should occur over weeks-months.
- **May continue** ULT indefinitely to reduce likelihood of future flares
- If a patient is already on ULT and experiences acute gout, do not stop the ULT → initiate acute treatment
- May initiate ULT during an acute attack or afterwards based on patient preference

PHARMACOLOGICAL MEASURES - CHRONIC PROPHYLAXIS

ALLOPURINOL (XANTHINE OXIDASE INHIBITOR)

- **1st line therapy**
- Minimum effective dose: 100-200 mg/day, maximum dose is 800 mg/day (divide doses if > 300 mg)
 - **Start at a low dose and titrate slowly to target serum levels < 360 µmol/L**
- Onset of efficacy: ↓ serum uric acid levels noted after days; however, overall clinical effect takes months
- **Side effects**: skin rash (maculopapular), diarrhea, nausea, vomiting, changes in liver function tests (e.g. ALT/AST), arthralgias, URTI, Allopurinol Hypersensitivity Syndrome (AHS)
 - **AHS** is a life-threatening rash presenting like Steven-Johnson's syndrome and toxic epidermal necrolysis; risk factors include starting at elevated doses, renal impairment, concomitant diuretic use, positive for HLA-B*5801 allele which is present mostly in Han Chinese, Thai, Korean, and African American patients
- **Renal dose adjustment required**
- **Interactions**:
 - Potentiates effects of **azathioprine** and **6-mercaptopurine** by ↑ half-life (↓ dose of 6-mercaptopurine or azathioprine by 75%)
 - **Thiazide** use potentiates allopurinol toxicity
 - Allopurinol + **ampicillin** increased the risk of a maculopapular rash
 - Allopurinol + **cyclophosphamide**: enhanced bone marrow suppression
 - Allopurinol + **probenecid**: probenecid reduces levels of allopurinol

PHARMACOLOGICAL MEASURES - CHRONIC PROPHYLAXIS

FEBUXOSTAT (XANTHINE OXIDASE INHIBITOR)

- Considered when allopurinol is not tolerated, ineffective, or if patients cannot use (e.g. risk factors for allopurinol hypersensitivity syndrome)
- Dose: 80 mg once daily
 - Start at a low dose and titrate slowly to target serum levels < 360 µmol/L
- Increases risk of cardiovascular-related deaths in patients with gout who have established cardiovascular disease; has a black box warning
- Side effects: nausea, vomiting, abdominal pain, dizziness, liver dysfunction
- Interactions
 - Potentiates effects of purine-based analogs (e.g. azathioprine or mercaptopurine) → leads to toxicity



Note: Other agents e.g. probenecid are no longer commonly used due to side effect profiles. Probenecid is available via the Special Access Program only in Canada

DRUGS THAT CAN IMPROVE URIC ACID LEVELS

DUE TO HAVING MILD URICOSURIC EFFECTS

Losartan

Increases urinary excretion of uric acid. Use if patient needs an anti-hypertensive agent and has gout; the only antihypertensive agent recommended by guidelines

Fenofibrate

Has urate lowering effects. Do not add or switch from statin to fenofibrate if patient has dyslipidemia as benefits do not outweigh risks

Amlodipine

Has urate lowering effects but not mentioned in guidelines



Note: Switching hydrochlorothiazide to an alternate antihypertensive is also recommended

MONITORING PARAMETERS

Relief of pain, redness and inflammation

Target serum uric acid level to $<360 \mu\text{mol/L}$
Levels should be monitored every 2-5 weeks until target is reached

Thereafter, serum urate levels should be checked every 3-6 months

Continue to monitor kidney function, hepatic enzymes, complete blood count, and electrolytes every 6-12 months

Patient response and tolerance

Side effects associated with drug therapy

TAKEAWAY

- Two main principles in gout management:
 1. Manage pain/inflammation
 2. Prevent future gout flare-ups
- **Avoid purine-rich foods** and mitigate metabolic syndrome risk factors
- Gout attacks will resolve without treatment within 3-10 days
- Pain and inflammation: NSAIDs, corticosteroids or colchicine are all good **first-line options**
 - Choice depends on patient-specific factors
- May start ULT during an acute gout episode
- Chronic prophylaxis with a ULT: allopurinol is **first-line**, duration likely indefinite, target serum urate $<360 \mu\text{mol/L}$
 - Febuxostat is another option
- **Patients should not stop allopurinol or febuxostat abruptly since it can precipitate an acute gout attack**
- **Monitor for blood dyscrasias and diarrhea with colchicine**

CASE

UA is a busy, active, 63-year-old male who is experiencing excruciating pain in his right big toe. UA had bloodwork done in the past and his uric acid level has been elevated for quite some time. This is the second time he has had these symptoms in the last year. He says this pain came on quickly and it is so bad that he cannot tolerate light pressure on the area. It has been 12 hours since his symptoms have started. His past medical history is significant for diabetes, hypertension and chronic kidney disease (CKD). He takes metformin 1000 mg po BID, liraglutide 1.2 mg SC daily and hydrochlorothiazide 25 mg po daily. He also takes acetaminophen 500 mg po prn for headaches. He is not allergic to anything. He drinks 2-3 alcoholic drinks per day and follows a high-protein, purine-rich diet.

Objective findings:

- Inflamed right big toe
- CrCl 26 mL/min 2 weeks ago

1. Which of the following factors is LEAST likely to have contributed to UA's gout?

- a) Purine-rich diet
- b) Diabetes
- c) Tylenol® (acetaminophen)
- d) Hydrochlorothiazide



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Objective findings:

- Inflamed right big toe
- CrCl 26 mL/min 2 weeks ago

2. Which of the following would be the best option to recommend for UA's acute gouty arthritis based on his medical history?

- a) Colchicine
- b) Ibuprofen
- c) Naproxen
- d) Prednisone



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Objective findings:

- Inflamed right big toe
- CrCl 26 mL/min 2 weeks ago

3. All of the following statements are correct EXCEPT:

- UA should be counselled on diet modification and lowering his alcohol intake
- Hydrochlorothiazide could be switched to losartan
- UA is likely to have more joints affected by gout in the future
- UA will not have any adverse effects if he starts ULT now



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Objective findings:

- Inflamed right big toe
- CrCl 26 mL/min 2 weeks ago

4. Which of the following would be most appropriate to start his urate-lowering therapy?

- a) Febuxostat
- b) Allopurinol
- c) Probenecid
- d) Losartan

Bonus Question

Can you spot any other clinical problem(s)?



REFERENCES

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CHANGE LOG

August 2025

- Minor formatting and grammatical changes made throughout
- Slide 18: Added 'antihypertensive' before agent
- Slide 24: Removed '50 mg daily' after losartan to match other options

November 2025

- Minor formatting changes throughout
- Slide 7: changed 'secretion' to 'excretion'

April 2026

- Slide 2: Updated NAPRA competencies
- Slide 6: Updated bullet point #3 for clarity
- Slide 7: Minor rephrasing regarding low dose ASA
- Slide 10: Edited and simplified information
- Slide 14: Minor updates and formatting changes for clarity
- Slides 16/17: Added concept of slow titration